

# BlinkIT Grocery Sales Analysis

## Business Requirements Document

### ***Prepared By***

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### ***Project Overview***

This project focuses on analyzing BlinkIT grocery sales data to understand sales performance, customer satisfaction, and product distribution patterns using SQL and Python analysis.

### ***Business Objective***

The objective is to identify key insights and optimization opportunities by analyzing sales trends, product categories, outlet performance, and customer ratings.

### ***Key Performance Indicators (KPIs)***

1. Total Sales: Overall revenue generated from all items sold.
2. Average Sales: Average revenue generated per sale.
3. Number of Items: Total count of items analyzed.
4. Average Rating: Average customer rating for items sold.

### ***Granular Analysis Requirements***

1. Total Sales by Fat Content

Objective: Analyze the impact of fat content on total sales.

2. Total Sales by Item Type

Objective: Identify the performance of different item types based on sales.

3. Fat Content by Outlet for Total Sales

Objective: Compare sales across outlets segmented by fat content.

4. Total Sales by Outlet Establishment

Objective: Evaluate how outlet establishment influences sales.

### ***Additional Analysis Requirements***

#### 5. Percentage of Sales by Outlet Size

Objective: Analyze the relationship between outlet size and total sales.

#### 6. Sales by Outlet Location

Objective: Analyze sales distribution across different outlet locations.

#### 7. All Metrics by Outlet Type

Objective: Compare Total Sales, Average Sales, Number of Items, and Average Rating across different outlet types.

### ***Technology Used***

SQL (PostgreSQL): Used for database creation, data cleaning, transformation, and business analysis queries.

Python: Used for data preprocessing, exploratory data analysis, and visualization using Pandas, NumPy, Matplotlib, and Seaborn libraries.

### ***Project Workflow***

Raw Dataset → Data Cleaning → PostgreSQL Database → SQL Queries → Python Analysis → Visualization → Business Insights

### ***Expected Outcome***

The project provides insights into product performance, outlet trends, customer satisfaction, and factors affecting BlinkIT sales performance.